1. Project Description

The **E-Certificate Automation Project** is a web-based system built using **AppSheet** and **Google Sheets**. Its goal is to replace manual paperwork with an automated digital workflow. The system allows employees to submit certificate requests for hardware or software acquisitions and renewals, which are then routed through a predefined chain of departmental approvers for digital signatures and final PDF generation.

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2. Access Roles and Permissions

To access the system, the app owner must share the app with specific roles. There are two primary roles defined in the AppSheet share settings, as seen in image 1.

Admin Role

- **Access Level:** "Edit app"
- **In-App Role:** "Admin"
- **App Version:** "Latest"
- **Permissions:** Admins can view and edit the **Approver_Settings** table to manage who approves requests for each department. They also have full visibility into the **Certificate_Requests** table, allowing them to see all requests made by every user in the organization.

User Role

- **Access Level:** "Use app"
- **In-App Role:** "User"
- **App Version:** "Latest"
- **Permissions:** Users cannot see or edit the **Approver_Settings** table. They can only access the **Certificate_Requests** table and are restricted to viewing only the requests they have personally submitted.

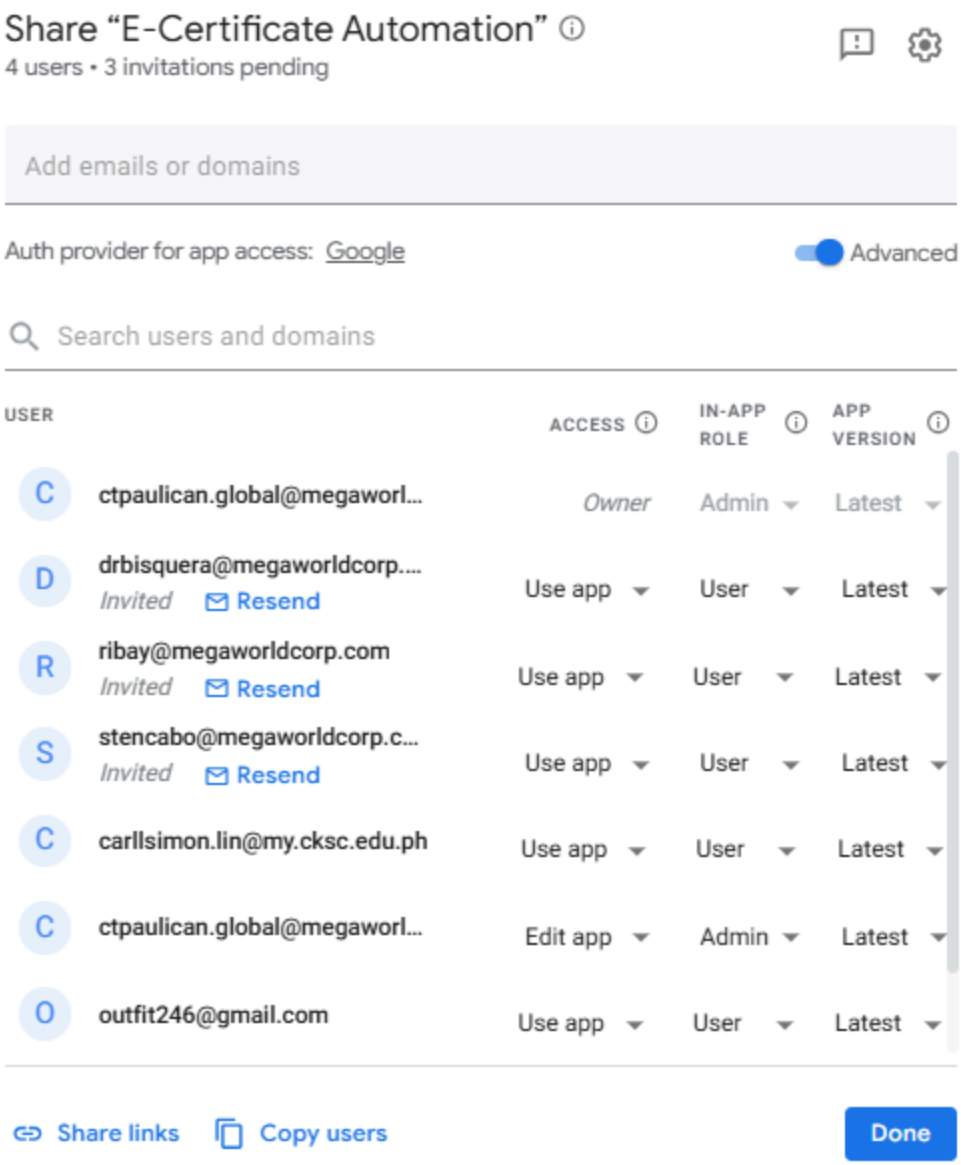


Image 1. AppSheet share settings

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3. Table Details

Approver_Settings Table

This table is managed solely by the Admin to define the approval hierarchy for different groups (departments).

- **How to add:** The Admin clicks the "Add" button to open a form.
- **Required Fields:** Group (Department), Approver 1 (Email), and Approver 1’s Designation

(Position).

- **Optional Fields:** Approver 2 (Email), Approver 2's Designation, Approver 3 (Email), and Approver 3's Designation.

- **Admin Control:** The Admin can add new settings or edit existing rows to update the approvers for a specific group.

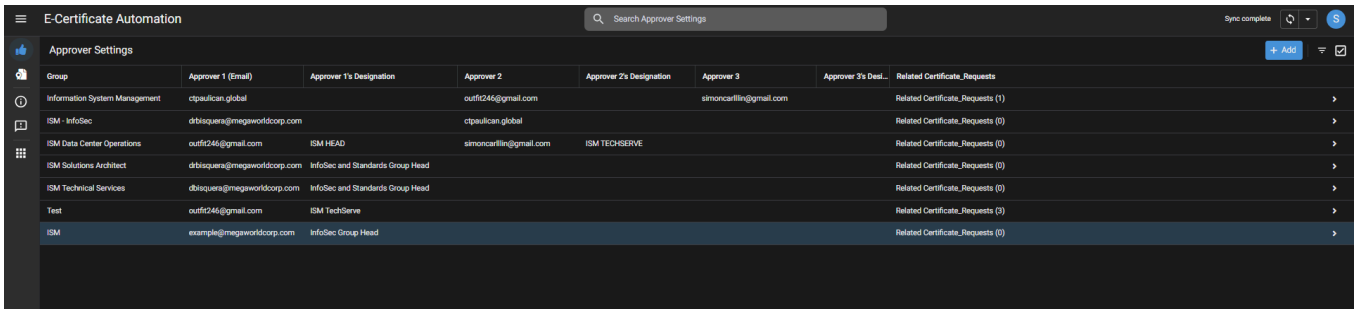


Image 2. Appearance of Approver Settings Table Database

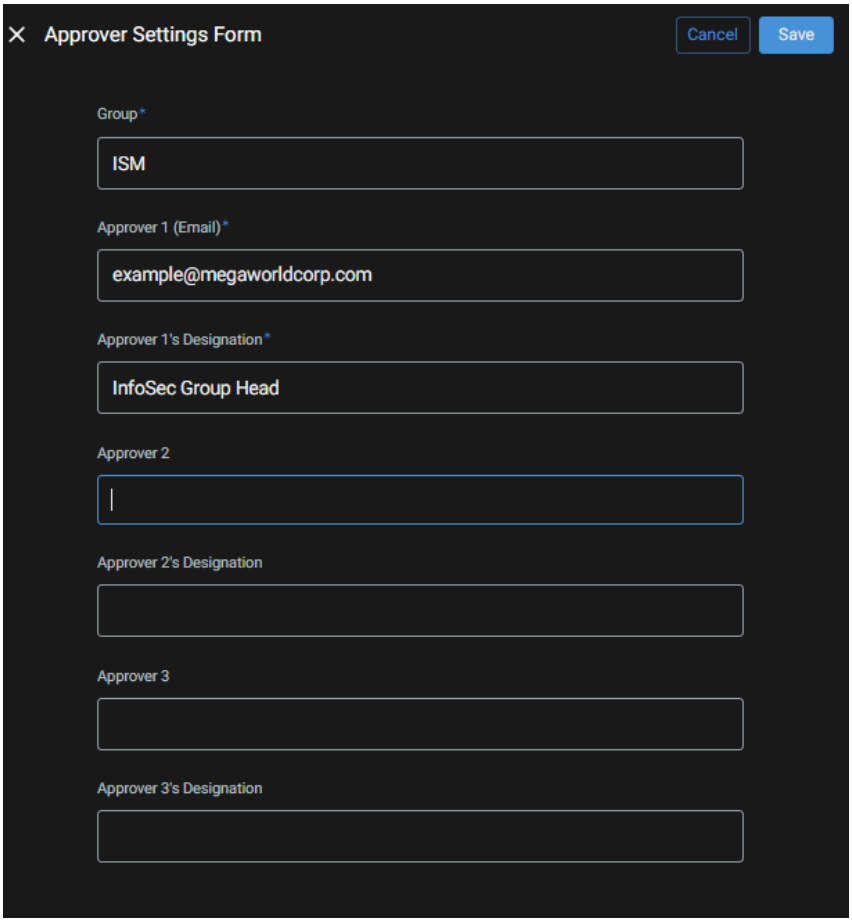


Image 3. Example of Approver Settings Form

Certificate_Requests Table

This is where the core request activity happens. Users create requests by clicking "Add," which generates a unique **Control Number**.

- **Request Form Fields:**
    - Control Number (Generates a unique ID) (i.e. EC-2026-00001-64f)
    - Project Title
    - Date of Filing (Calendar picker)
    - Requestor (The email of the requestor is shown)
    - Requestor's Designation (Input the Requestor's Position)
    - Group (Target Department)
    - Accountable Person and their Department
    - Location (Options: AGT, GCP, EGP, TWS, or "Other" by adding their own location)
    - PC Name, Purchase Order (PO), PRISM No, and iHelp Number.
    - Product Type (Hardware or Software) and Product Name.
    - Type of Purchase (New Acquisition, Renewal, or Replacement).
    - Type of License
    - Contract Period (Start and End Dates).
    - Supplier Name
    - Requestor's Remark (Purpose of request).
    - **Supporting Documents:** Users can attach multiple files by clicking "New", which a Request Attachment Form appears where users can upload the file, and provide an "Attachment Label" description.

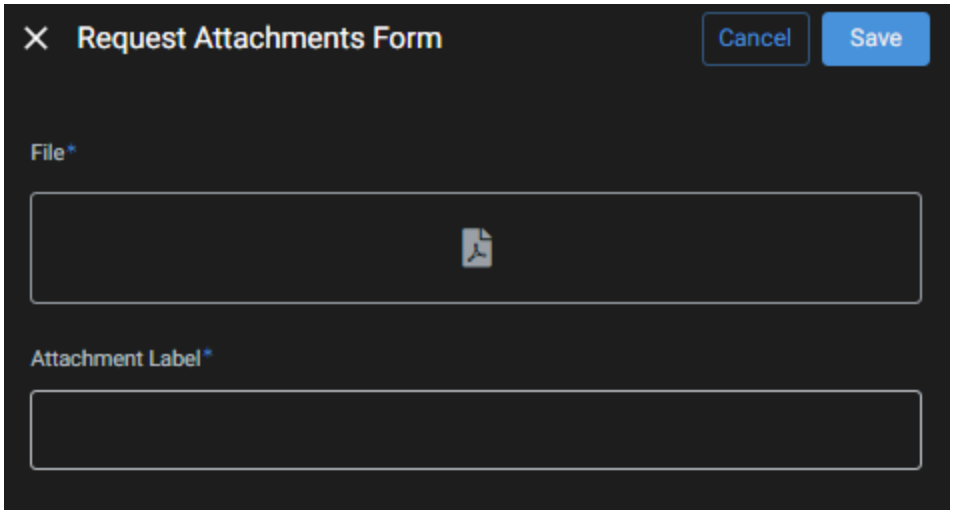
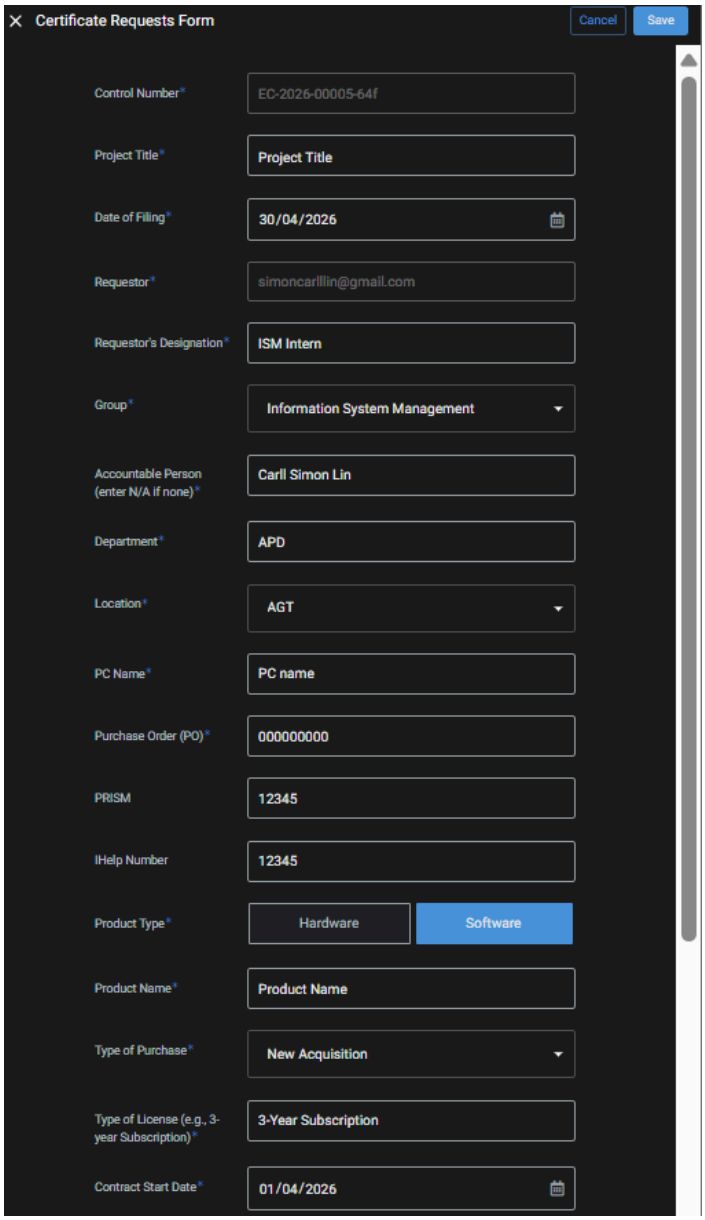


Image 4. Request Attachments Form


Certificate Requests Form
Cancel
Save
Control Number*
EC-2026-00005-64f
Project Title*
Project Title
Date of Filing*
30/04/2026
Requestor*
simoncarlllin@gmail.com
Requestor's Designation*
ISM Intern
Group*
Information System Management
Accountable Person (enter N/A if none)*
Carll Simon Lin
Department*
APD
Location*
AGT
PC Name*
PC name
Purchase Order (PO)*
000000000
PRISM
12345
IHelp Number
12345
Product Type*
Hardware
Software
Product Name*
Product Name
Type of Purchase*
New Acquisition
Type of License (e.g., 3-year Subscription)*
3-Year Subscription
Contract Start Date*
01/04/2026

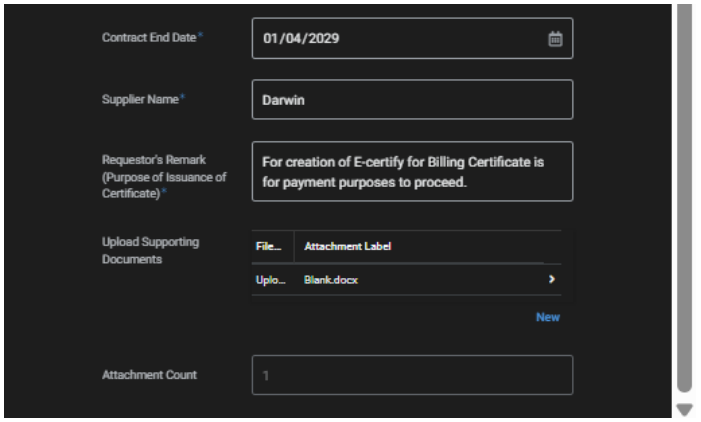

Contract End Date*
01/04/2029
Supplier Name*
Darwin
Requestor's Remark (Purpose of Issuance of Certificate)*
For creation of E-certify for Billing Certificate is for payment purposes to proceed.
Upload Supporting Documents
File...
Attachment Label
Uplo...
Blank.docx
New
Attachment Count
1

Image 5. Example of Certificate Requests Form

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4. How the Workflow Works

Step 1: Submission

Once the user saves the request form, the system automatically identifies the **Approver 1** assigned to that specific group and sends them an email notification.

Step 2: Approval Actions

The approver receives a digital card containing all request details and three action buttons:

1. **Approve:** Moves the request to the next approver in the chain (Approver 2, then Approver 3).

2. **Decline:** Stops the request immediately. **Approver Remark is mandatory.** The system notifies the requestor that the request is void and they must start a new one.

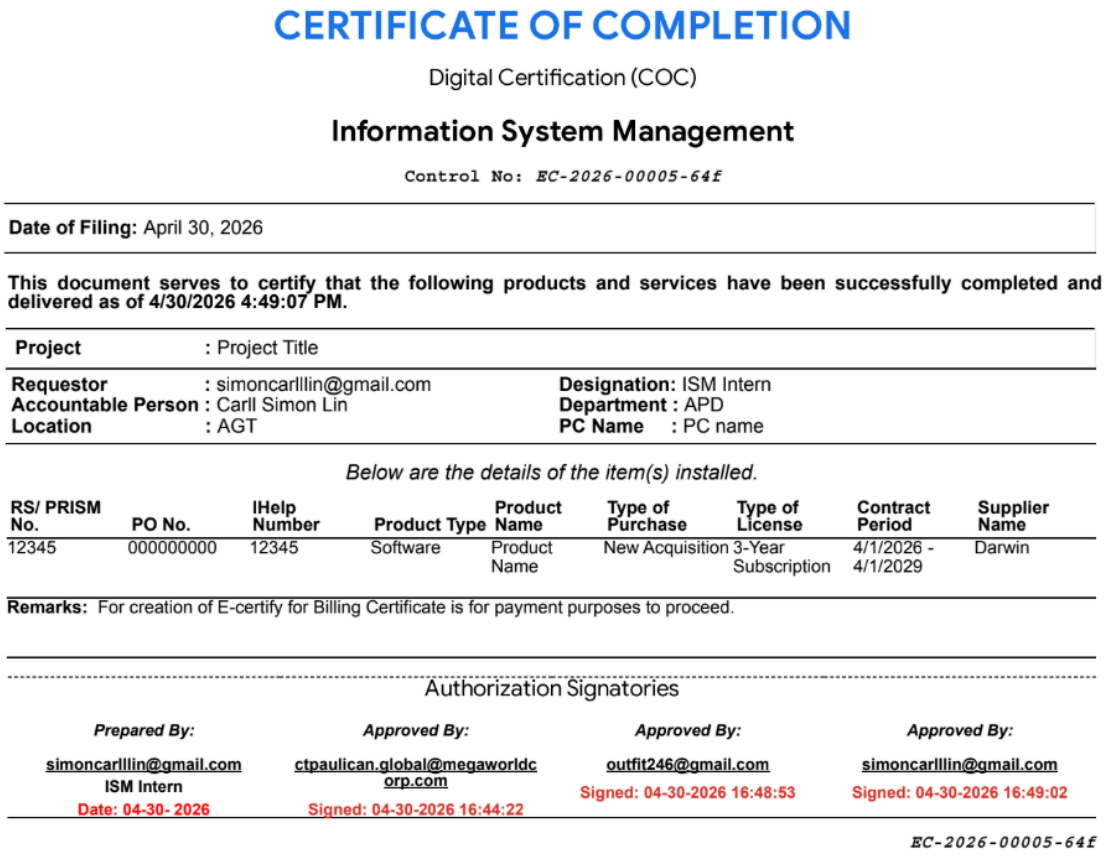
3. **Return Request:** Sends the request back to the user for corrections. **Approver Remark is mandatory.** Once the user resubmits, it goes back to Approver 1.

Step 3: Confirmations

Every time an approver clicks Approve, Decline, or Return, the system sends a confirmation email to that specific approver to verify their action was successful.

Step 4: Final Completion

When the final required approver provides their approval, the system changes the status to "Fully Approved." It then automatically generates a **Certificate of Completion** PDF and emails it directly to the requestor as an attachment.



CERTIFICATE OF COMPLETION

Digital Certification (COC)

Information System Management

Control No: EC-2026-00005-64f

Date of Filing: April 30, 2026

This document serves to certify that the following products and services have been successfully completed and delivered as of 4/30/2026 4:49:07 PM.

Project	: Project Title		
Requestor	: simoncarlllin@gmail.com	**Designation:**	ISM Intern
Accountable Person	: Carll Simon Lin	**Department :**	APD
Location	: AGT	**PC Name**	: PC name

Below are the details of the item(s) installed.

RS/ PRISM No.	PO No.	IHelp Number	Product Type	Product Name	Type of Purchase	Type of License	Contract Period	Supplier Name
12345	000000000	12345	Software	Product Name	New Acquisition	3-Year Subscription	4/1/2026 - 4/1/2029	Darwin

Remarks: For creation of E-certify for Billing Certificate is for payment purposes to proceed.

Authorization Signatories

Prepared By:	*Approved By:*	*Approved By:*	*Approved By:*
simoncarlllin@gmail.com ISM Intern Date: 04-30- 2026	ctpaulican.global@megaworldcorp.com Signed: 04-30-2026 16:44:22	outfit246@gmail.com Signed: 04-30-2026 16:48:53	simoncarlllin@gmail.com Signed: 04-30-2026 16:49:02

EC-2026-00005-64f

Image 6. Example of Certificate of Completion

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5. Project Links

- **E-Certification Automation App (Browser View):**
  https://www.appsheet.com/start/90f04812-4642-4c6b-b671-b738b8ef3489

- **E-Certification Automation App (Editor/Admin View):**
  https://www.appsheet.com/Template/AppDef?appName=E-CertificateAutomation-864848063

- **Google Sheet Database:**
  https://docs.google.com/spreadsheets/d/12-o9ERCer1kPSEM8B0oaeKVk8XV3JNtSO8fu54kamS4/edit?usp=sharing

- **Google Drive Folder (Attachments & PDFs):**
  https://drive.google.com/drive/folders/1RJ1P7vZXgLa85KfsdQKCLBLqsaDqde-Y?usp=drive_link